



HWA CHONG INSTITUTION
C1 Promotional Examination
Higher 2

NAME

CT GROUP

19S

CHEMISTRY

9729/02

Paper 2 Structured questions

2 October 2019

1 hour

Candidates answer on the Question Paper.

Additional Materials: Data Booklet.

READ THESE INSTRUCTIONS FIRST

Write your name and CT group on all the work you hand in.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

A Data Booklet is provided.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

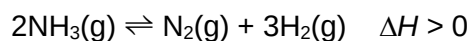
FOR EXAMINERS' USE ONLY

Paper 1 (25.4%)	Paper 2 (35.6%)	Paper 3 (39.0%)	100 %
Multiple Choice	Structured	Free Response	
	Q1 / 9	Q1 / 20	
	Q2 / 8	Q2 / 3 / 15	
	Q3 / 8		
	Q4 / 10		
	Deductions	Deductions	
/ 20	Subtotal: / 35	Subtotal: / 35	

Answer **all** the questions.

- 1** Ammonia decomposes spontaneously in the presence of iridium metal catalyst into nitrogen and hydrogen according to the equation below.

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- (a)** State two ways in which the properties of an ideal gas differs from a real gas.

.....

.....

.....[2]

- (b) (i)** A sample of ammonia was found to be 15% decomposed in the presence of the iridium metal catalyst. The partial pressure of nitrogen in the mixture was found to be 10 kPa.

Calculate the partial pressure of ammonia remaining in the mixture.

[1]

- (ii)** Hence or otherwise, calculate the total pressure of the mixture after decomposition.

[1]

- (iii) Describe the mode of action of the iridium catalyst in the decomposition of ammonia.

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.....[3]

- (c) Deduce the effect on the spontaneity of the decomposition of ammonia when the temperature is increased.

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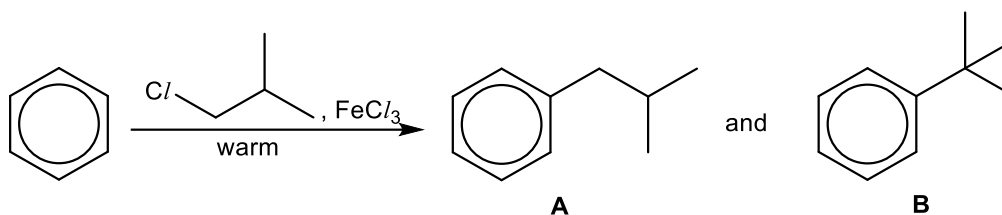
.....

.....[2]

[Total: 9]

- 2 Benzene is often used as a starting material to make other chemicals. Compounds **A** and **B** can be synthesised from benzene by the following reaction.

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- (a) (i) Draw the structure of the electrophile that results in the formation of **A**.

[1]

- (ii) Write an equation to show how the electrophile in (a)(i) is generated.

[1]

- (iii) Hence, explain why FeCl_3 acts as a Lewis acid in the reaction.

.....

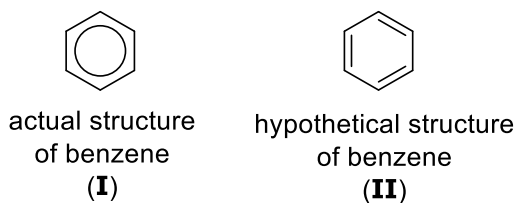
[1]

- (iv) Suggest the reagents and conditions you would use to distinguish **A** and **B**. State the observations you would expect for each compound.

.....

[2]

- (b) The actual structure of benzene (**I**) is 149 kJ mol^{-1} more stable than the hypothetical structure of benzene (**II**).

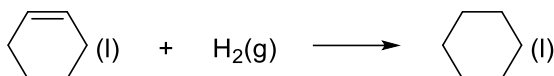


- (i) Explain why (**I**) is more stable than (**II**).

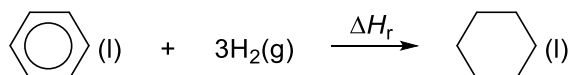
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[1]

The enthalpy change of hydrogenation for cyclohexene to cyclohexane is -118 kJ mol^{-1} .



Under the right conditions, benzene can also undergo hydrogenation to form cyclohexane.



- (ii) By drawing a suitable energy level diagram and incorporating the data given above, calculate the enthalpy change of hydrogenation of benzene, ΔH_r .

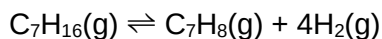
energy/ kJ mol^{-1}



enthalpy change of hydrogenation of benzene, $\Delta H_r =$

[2]
[Total: 8]

- 3** High octane fuels that are free from lead additives often contain aromatic hydrocarbons such as methylbenzene, C_7H_8 , which can be obtained from alkanes by the process of “reforming”. In a typical plant, heptane vapour is mixed with an inert carrier gas and passed over a catalyst at 300 °C.



- (a) (i)** Write the K_p expression for this reaction.

.....[1]

- (ii)** When 10 mol of heptane was passed over the catalyst at 300 °C and allowed to reach equilibrium, it was found that the mixture contained 8 mol of hydrogen and the total pressure was 3 atm.

Calculate the equilibrium constant K_p for this reaction, giving the units.

[3]

- (iii)** Explain how the yield of methylbenzene would be affected when a smaller vessel is used for the same reaction at 300 °C.

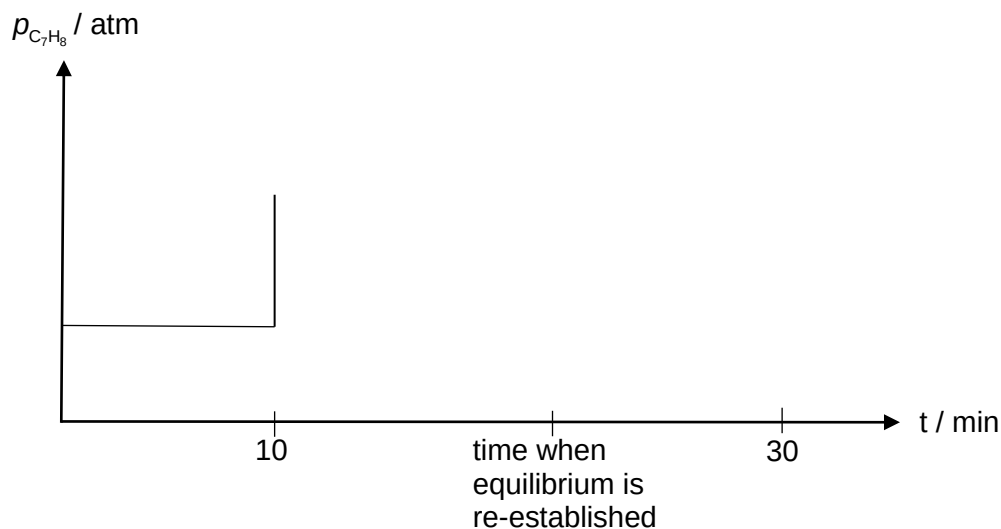
.....

[1]

- (iv) The system in (a)(ii) was left at equilibrium for 10 min. At $t = 10$ min, some C_7H_8 was added and the system was allowed to re-establish equilibrium.

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Using the given axes, sketch the graph for the variation of the partial pressure of C_7H_8 , $p_{C_7H_8}$, from $t = 10$ min to $t = 30$ min.



[1]

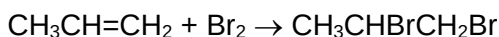
- (b) A mixture of 10 cm^3 of methane and 20 cm^3 of an unknown gas C_xH_y was completely combusted with 100 cm^3 of oxygen. After cooling to room temperature, 70 cm^3 of the gas mixture remained. This was passed through $Ca(OH)_2(aq)$ and the volume decreased to 20 cm^3 .

Find the molecular formula of the unknown gas.

[2]

[Total: 8]

- 4 Propene reacts with bromine in tetrachloromethane solvent to form 1,2-dibromopropane according to the following equation:



To investigate the kinetics of this reaction, a series of experiments was carried out using different volumes of propene solution, bromine solution and tetrachloromethane. The time taken for the disappearance of the colour of bromine, when all the bromine was completely reacted, was recorded.

The following results were obtained.

experiment	volume of propene solution/ cm ³	volume of bromine solution/ cm ³	volume of tetrachloromethane/ cm ³	time taken for disappearance of colour of bromine/ s
1	20	20	0	18
2	12	20	8	30
3	20	10	10	18

- (a) Explain why different volumes of tetrachloromethane were added to each set of experiment.

.....
[1]

- (b) What is the simple relationship between the time taken for disappearance of the colour of bromine and the rate of reaction?

.....[1]

- (c) Use the data in the table to deduce the order of reaction with respect to propene.

Order with respect to propene

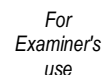
[2]

- (d) Use the data in the table to explain why the order of reaction with respect to bromine is one.

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[1]

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